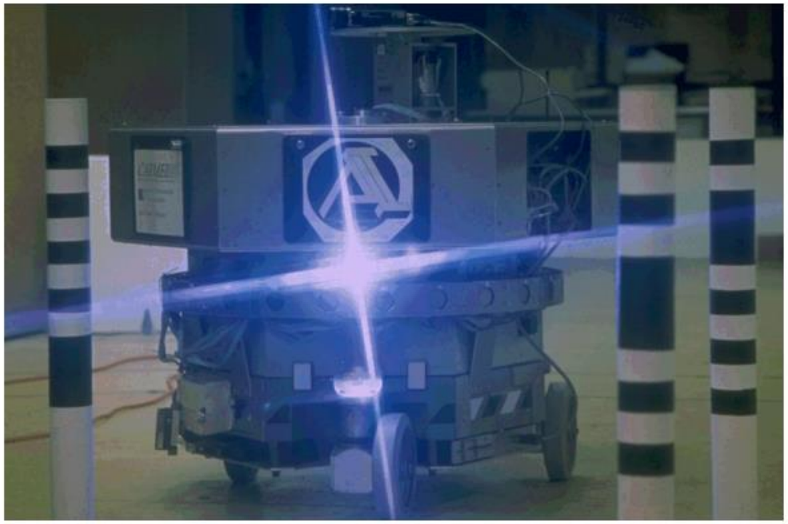
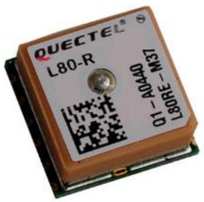
# Sensors for mobile robot

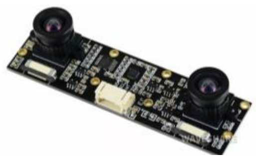


# Sensors for Robots- Mobile robot

**GPS**




**Deep camera**



**Lidar sensor**


**Line sensor**


**Approximate sensor**

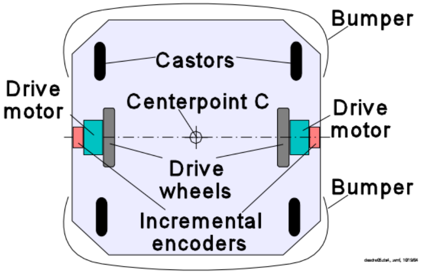
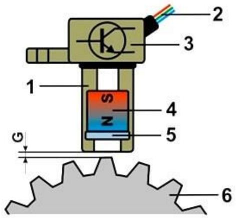
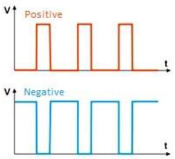


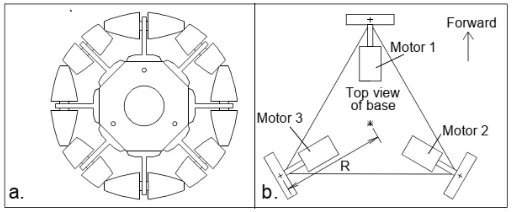

**Figure 1.6:** A typical differential-drive mobile robot (bottom view).

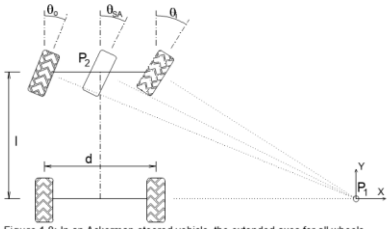
**Motor: Encoder – Hall sensors**




**IMU sensor**



# ROS and Sensors

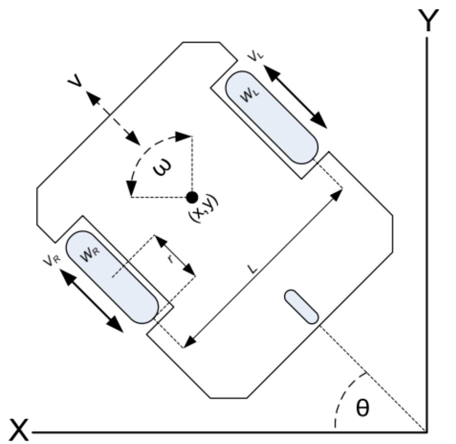



**Fig. 2:** The kinematics model of experimental DMR

$$V_R = \frac{v - \frac{L \times \omega}{2}}{r}$$

$$V_L = \frac{v - \frac{L \times \omega}{2}}{r}$$

$$\dot{x} == \frac{r}{2}(V_R + V_L)\cos\theta$$

$$\dot{y} == \frac{r}{2}(V_R + V_L)\sin\theta$$

$$\dot{\theta} = \frac{r}{L}(V_R + V_L)$$

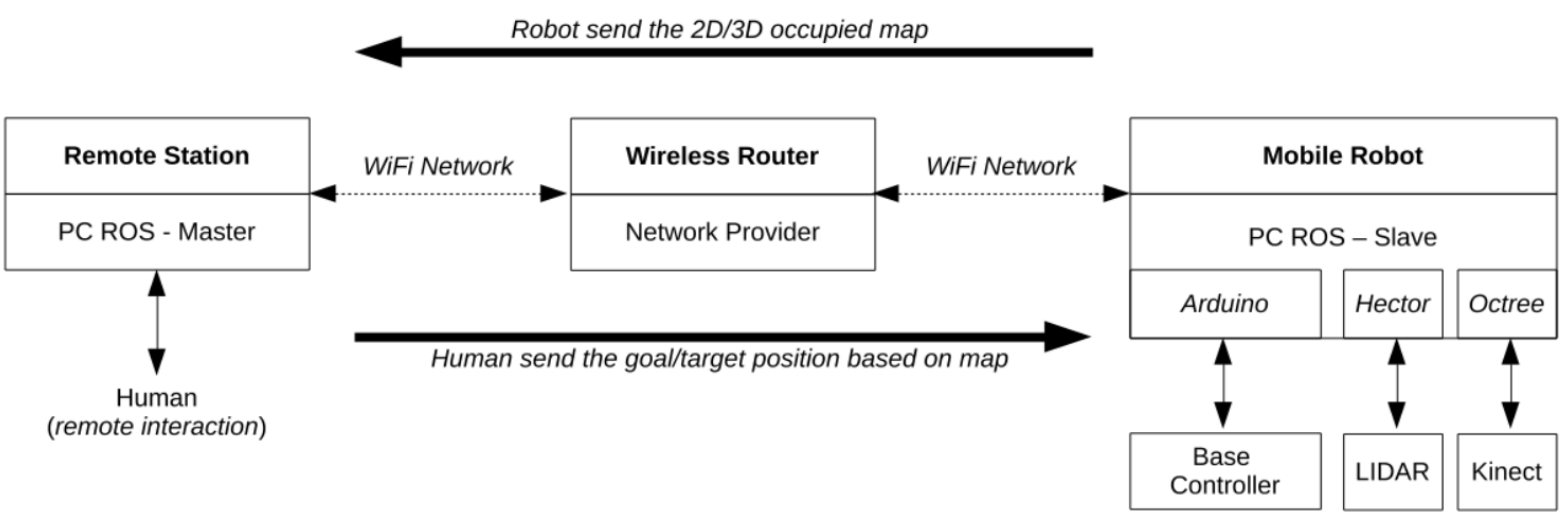



**Fig. 1:** The architecture of proposed system based on ROS framework

# ROS - Sensors

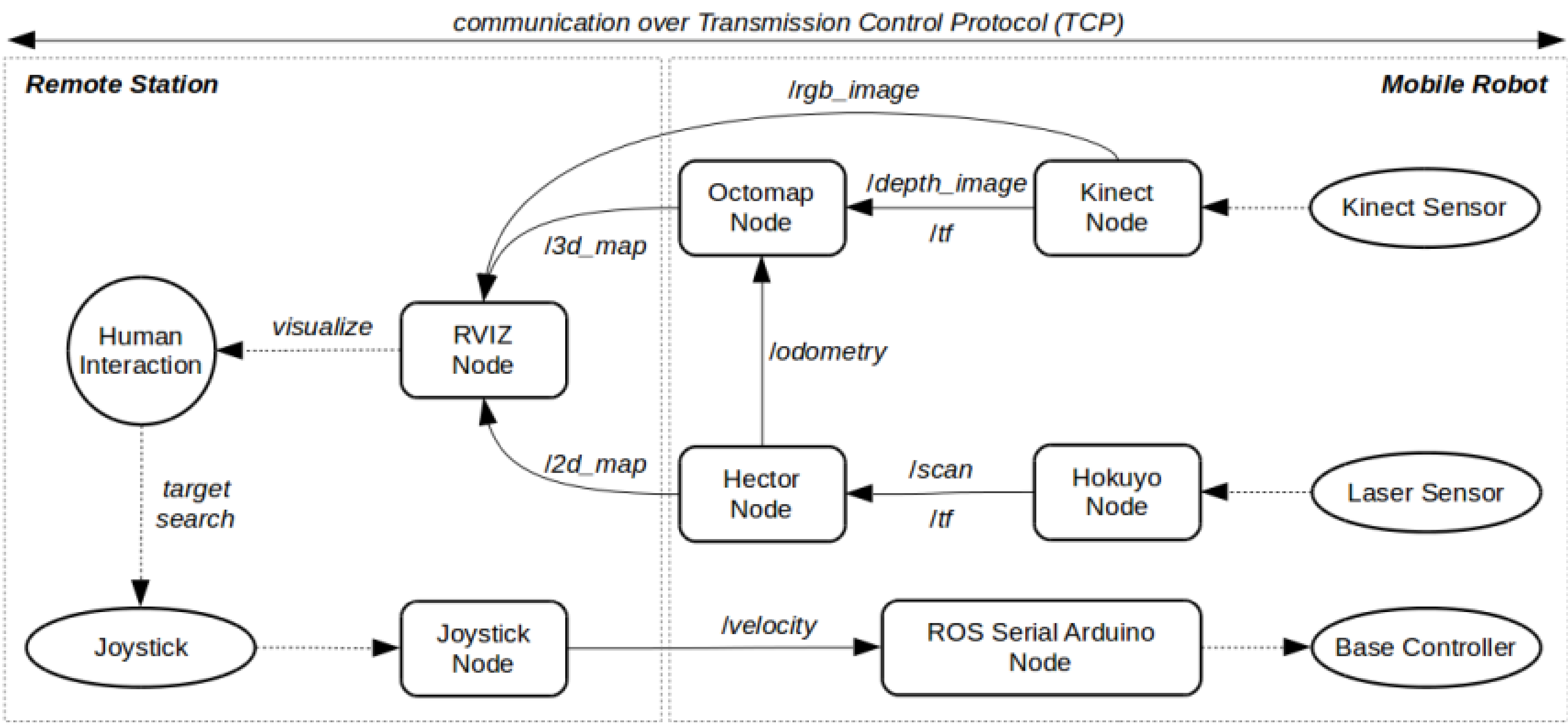



**Fig. 6:** Experimental ROS ecosystems configuration

# Laser scanner- LIDAR

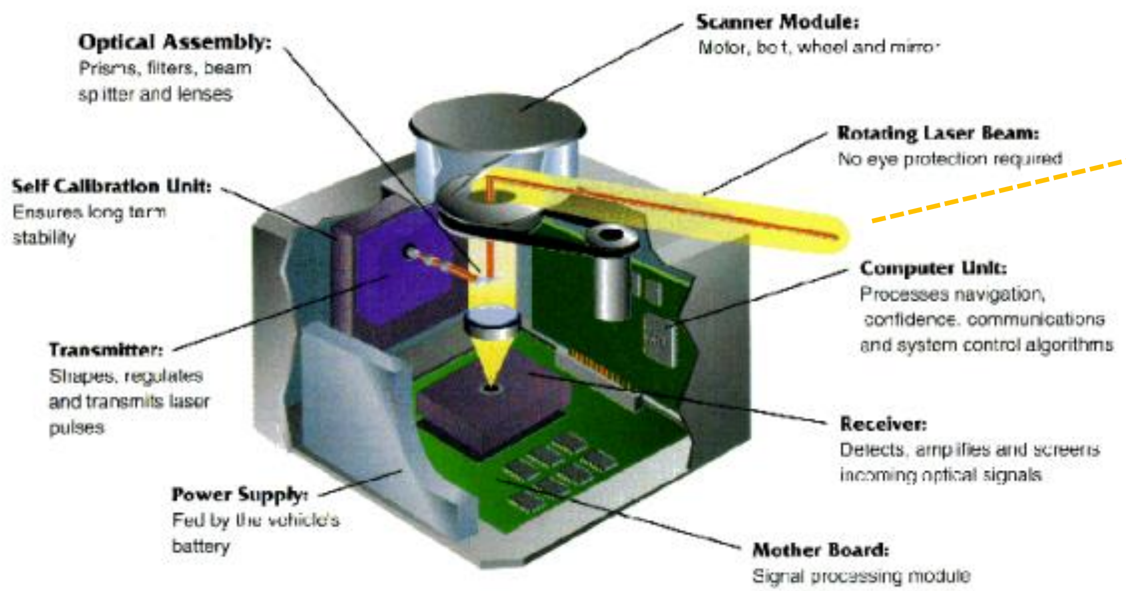
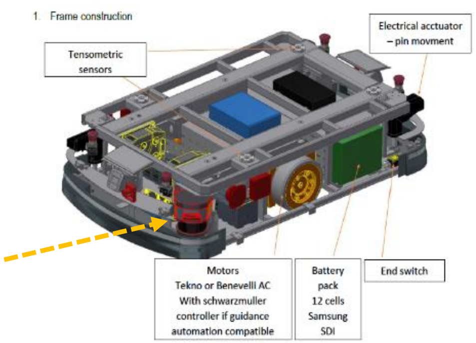



**Figure 6.17:** The *ROBOSENSE* scanning laser rangefinder was developed by Siman Sensors & Intelligent Machines Ltd., Misgav, Israel. The system determines its own heading and absolute position with an accuracy of 0.17° and 20 millimeters (3/4 in), respectively. (Courtesy of Siman Sensors & Intelligent Machines.)



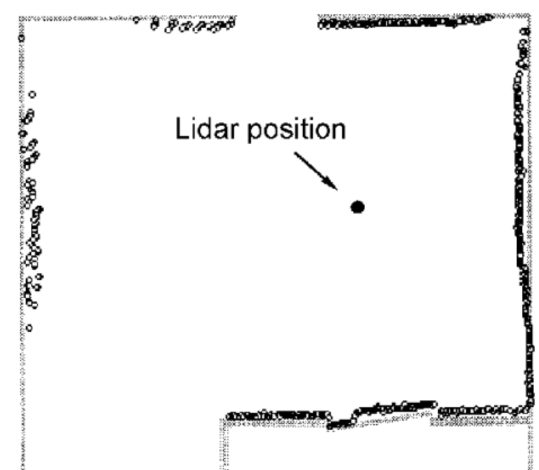
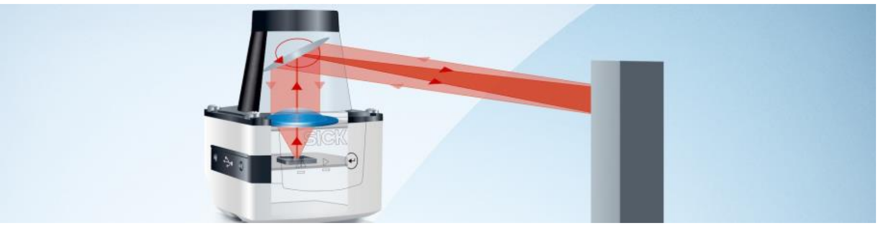




**Figure 8.4:** A typical scan of a room, produced by the University of Kaiserslautern's in-house developed lidar system. (Courtesy of the University of Kaiserslautern.)

# Laser scanner- LIDAR



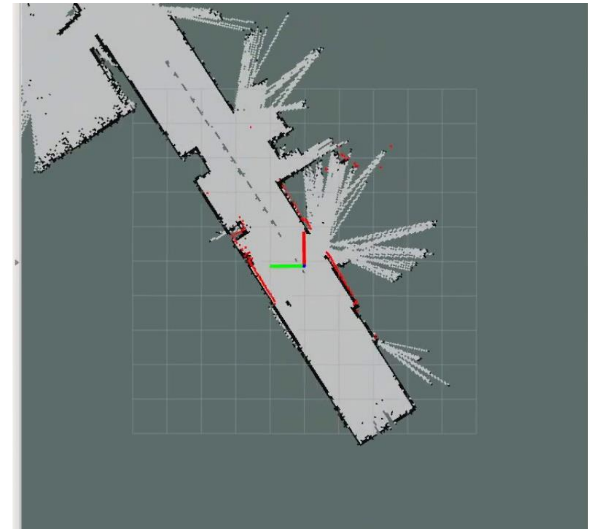

UK
CA

Technical data overview

| | |
|---|---|
| **Application** | Indoor / Indoor / Outdoor (depending on type) |
| **Measurement principle** | HDDM+ |
| **Integrated application** | Output of measurement data, 2D Object Detection, 2D Object Detection Advanced, LiDAR-LOC 2, Virtual Line Navigation, CODE-LOC (depending on type) |
| **Aperture angle** | |
| Horizontal | 276° |
| **Working range** | 0.05 m ... 120 m (depending on type) |
| **Scanning range** | |
| At 90% remission and 10 klx | 25 m ... 75 m (depending on type) |
| At 10% remission and 10 klx | 12 m ... 40 m (depending on type) |
| **Amount of evaluated echoes** | 1 / 3 (depending on type) |
| **Scanning frequency** | 20 Hz, 15 Hz, 25 Hz, Depends on the Dynamic Sensing Profile, 30 Hz, 40 Hz, 50 Hz (depending on type) |
| **Ambient operating temperature** | -33 °C ... +50 °C (depending on type) |
| **Ethernet** | ✓ |
| **Weight** | 220 g / 250 g / 220 g (depending on type) |

Product description

2D Lidar



3D Lidar

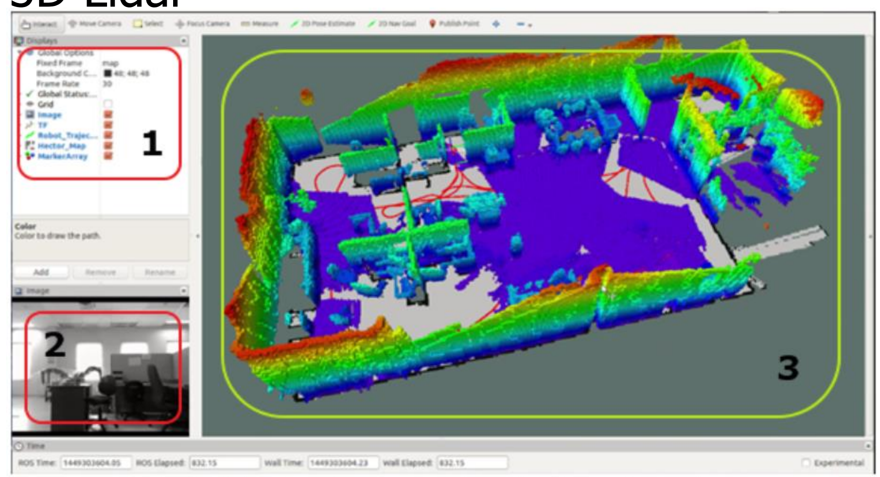
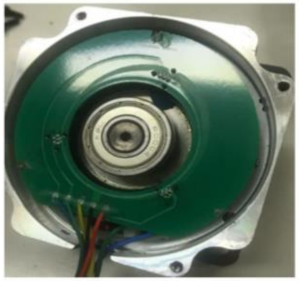


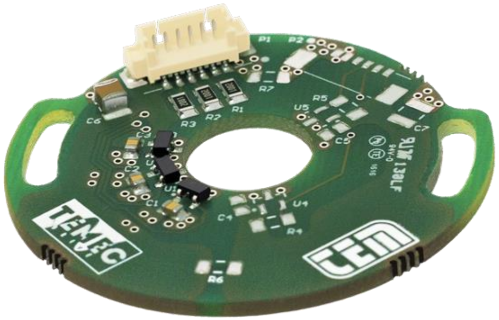

**Fig. 7:** RVIZ node panel for human-robot visual interface on ROS ecosystem

# BLDC MOTOR

ROBOT Dynamic control



BLDC với cảm biến Hall được lắp ở khoảng cách 120°.



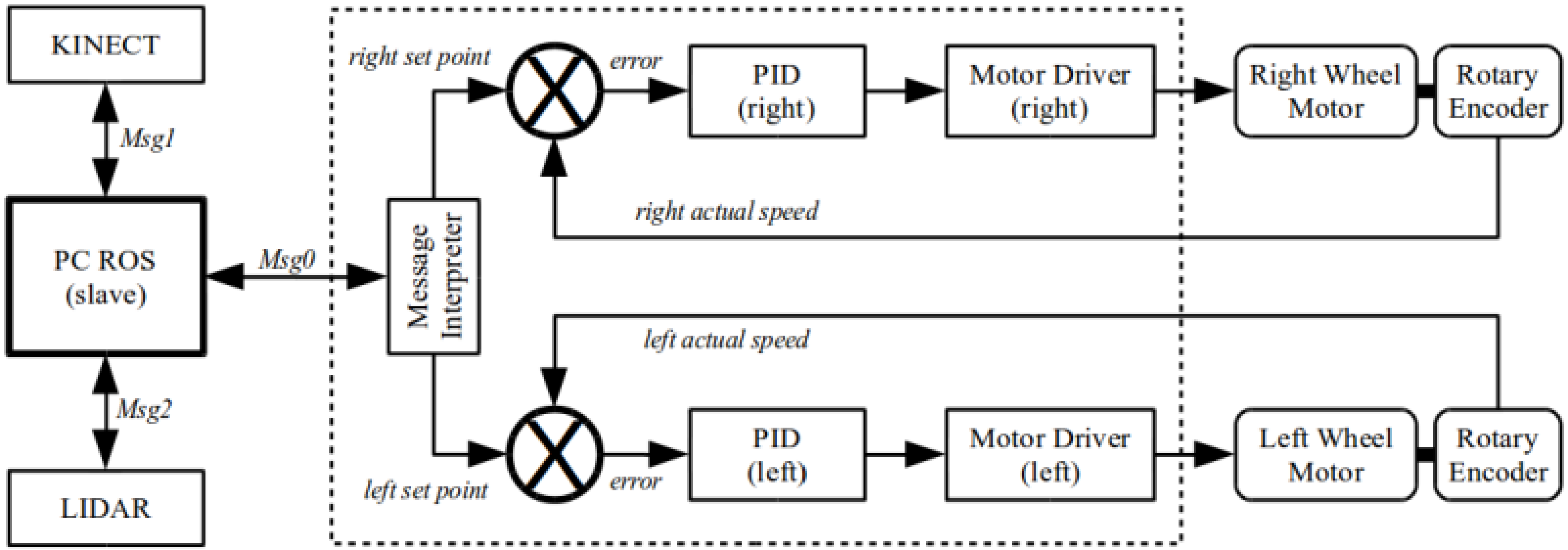
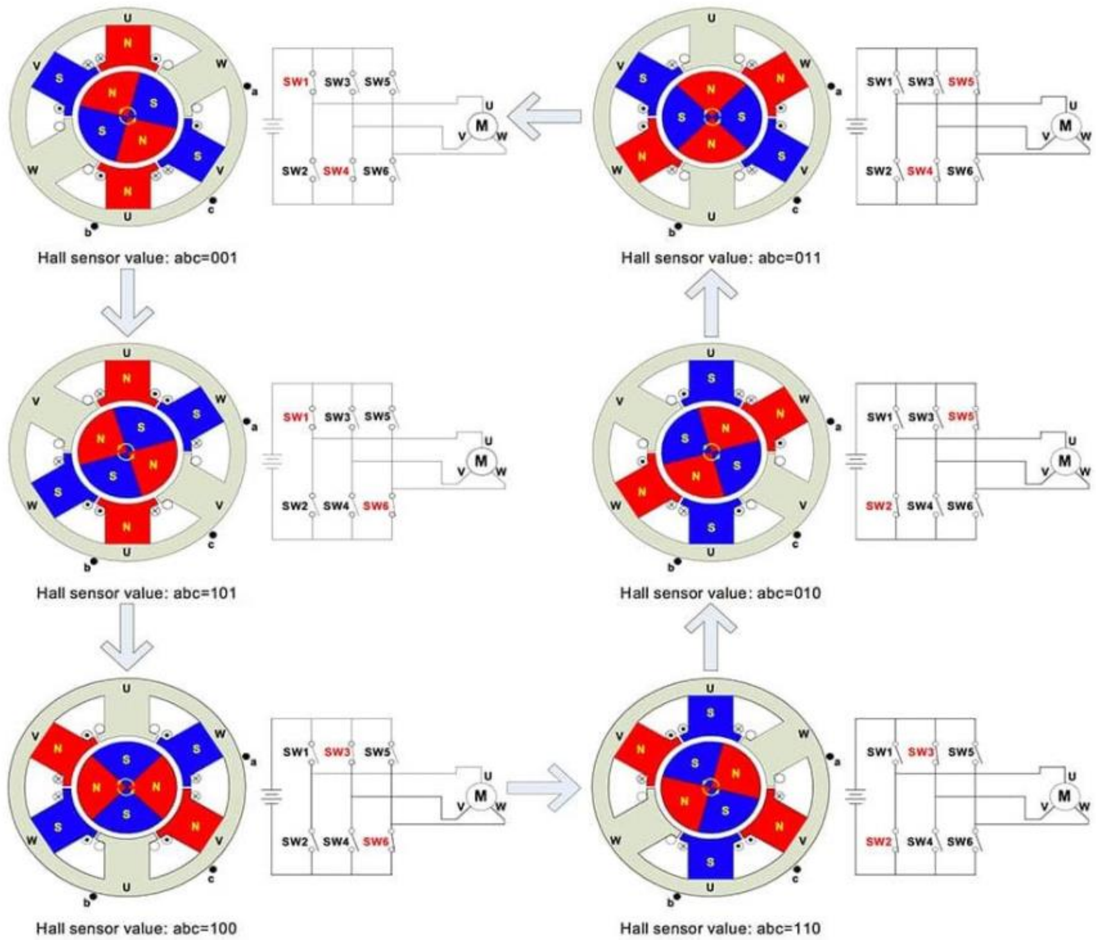



**Fig. 3:** The configuration of controller for DMR

# ROBOT-BLDC MOTOR

## BLDC MOTOR



# ROBOT-BLDC MOTOR

## BLDC MOTOR

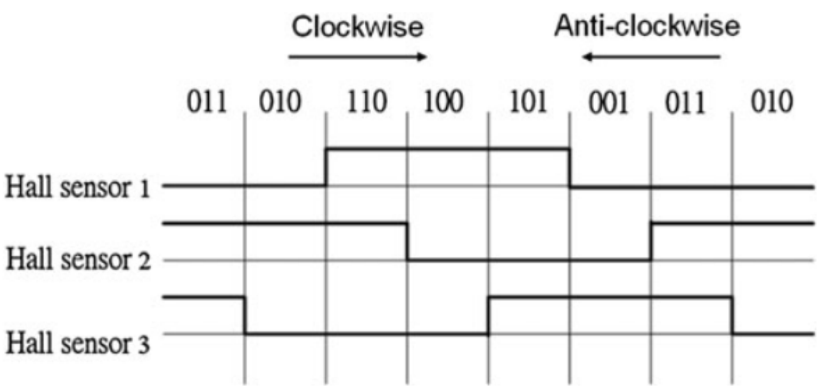
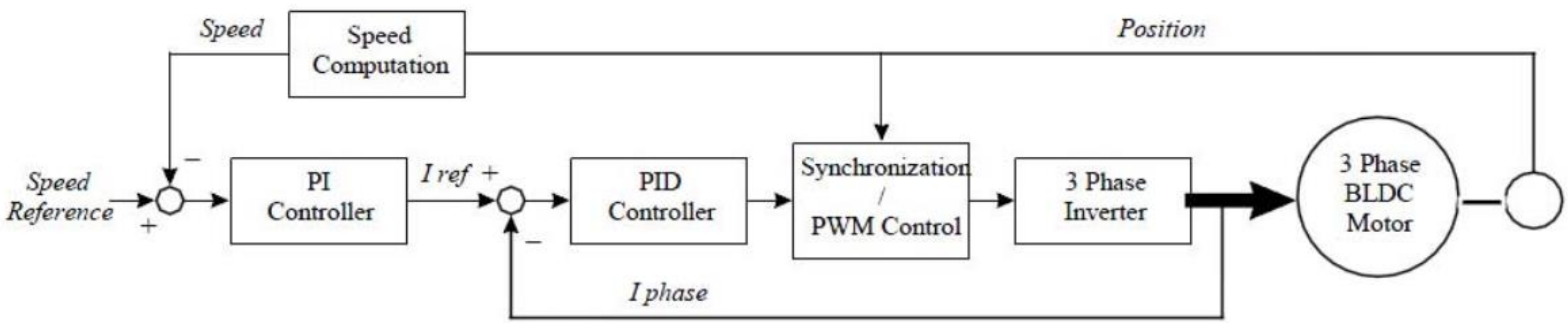


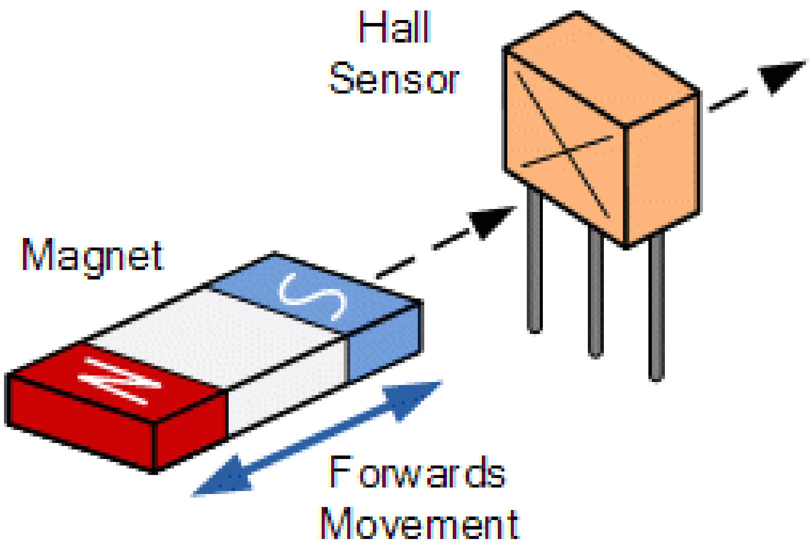

**Fig. 3** The intervals of the Hall sensor updating

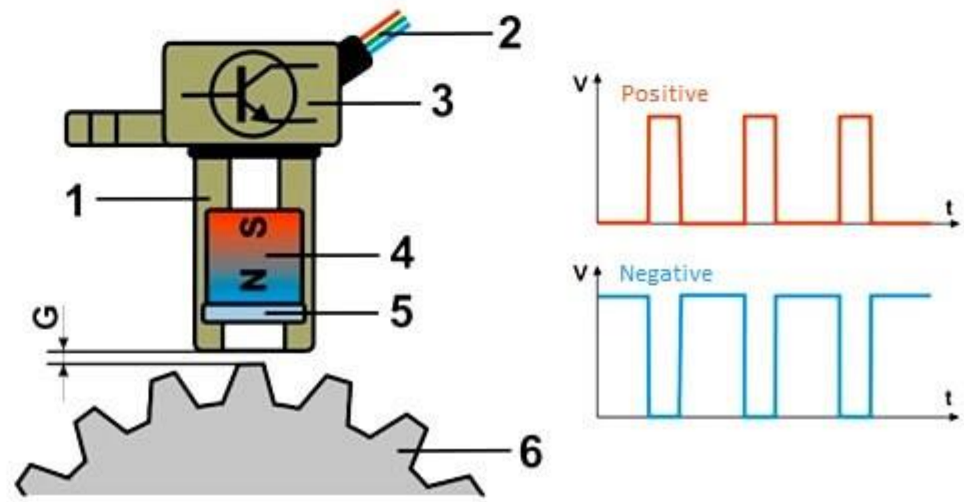
| a | b | c | HALL1 | HALL2 | HALL3 |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 |



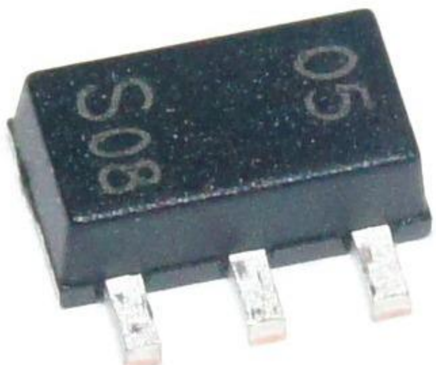

(a)

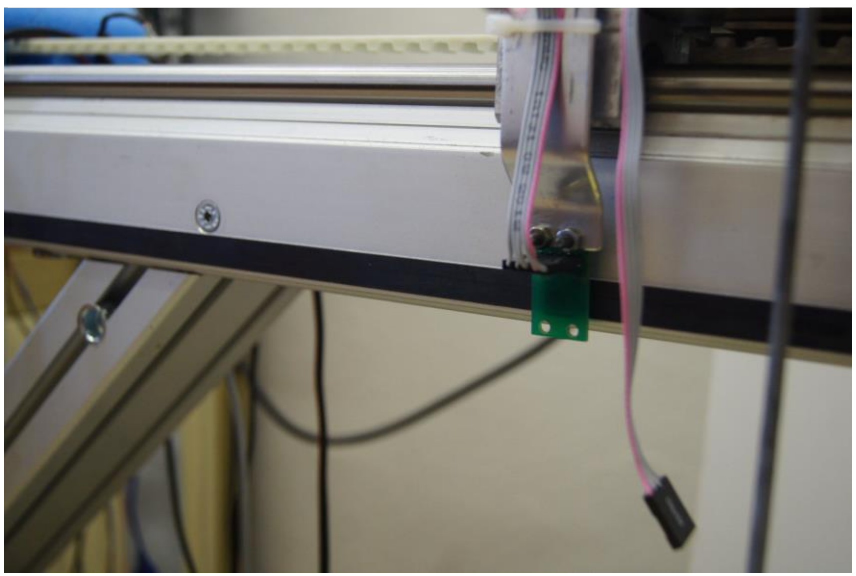
# Magnetic (Hall) sensors

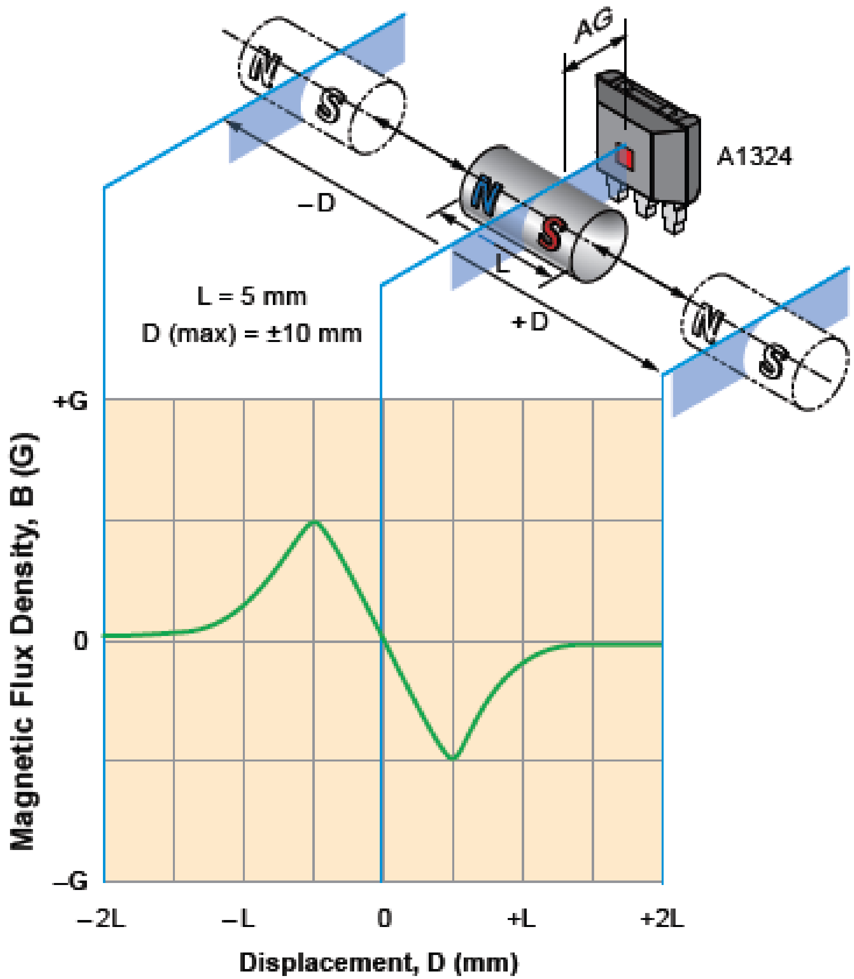


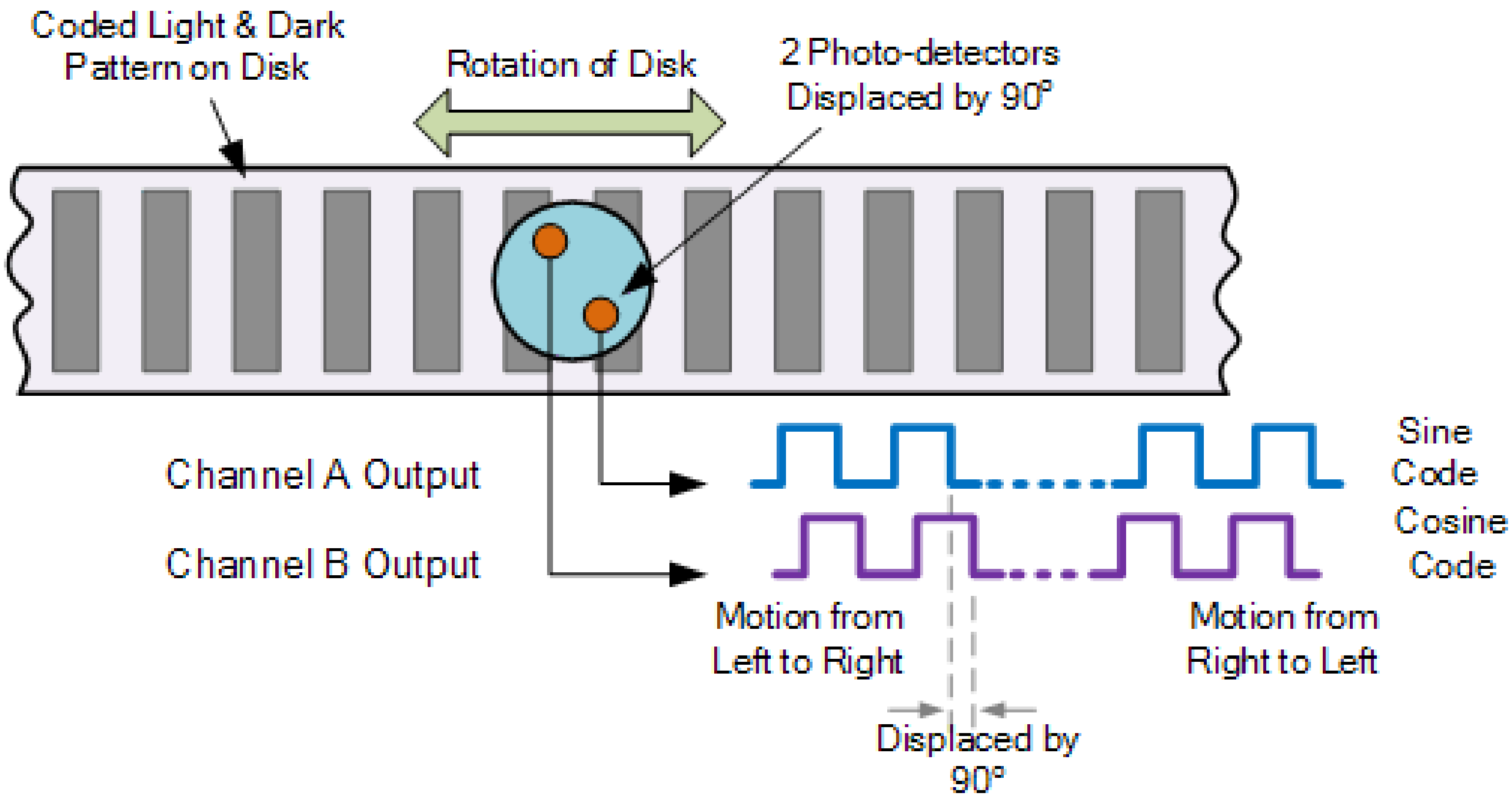
# Magnetic (Hall) sensors



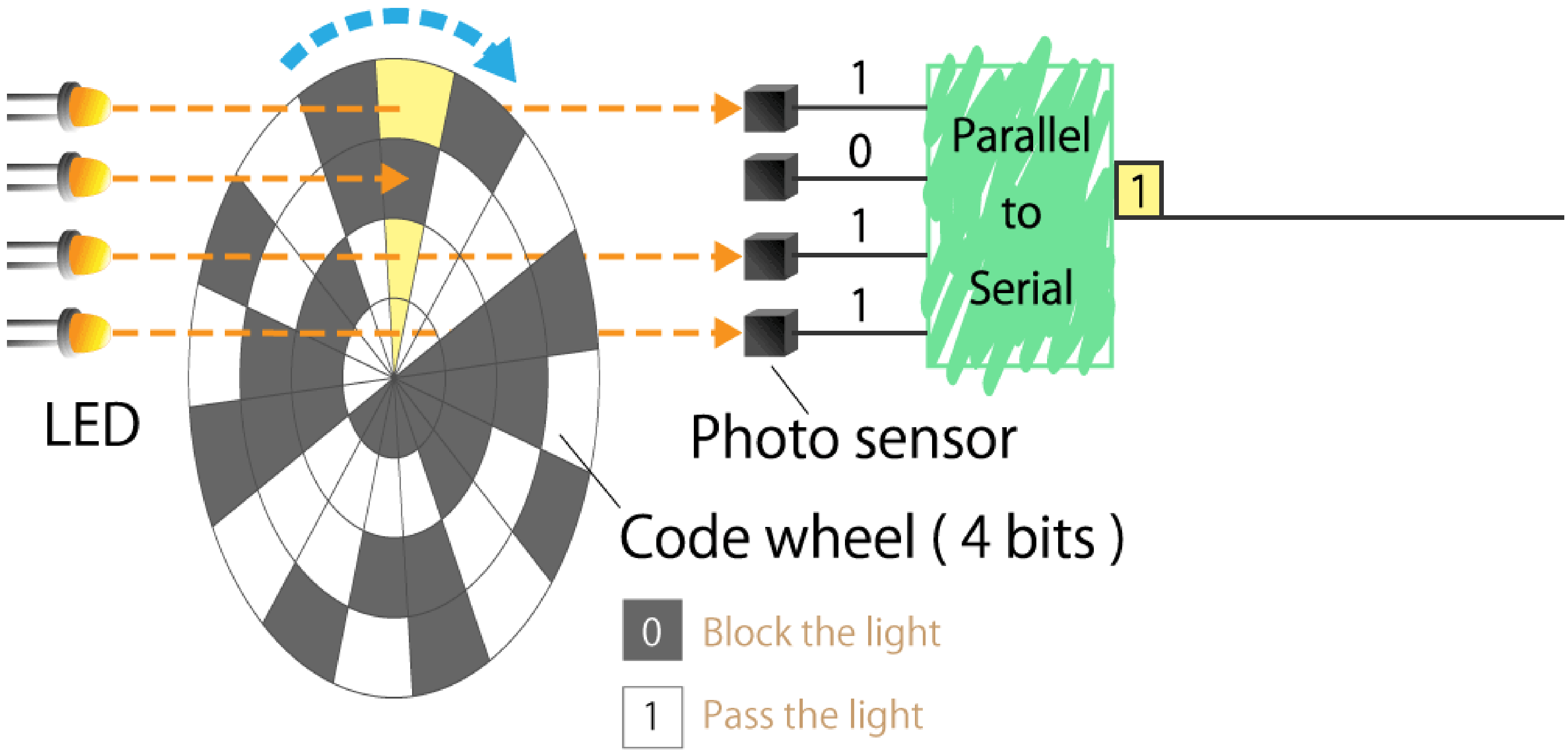





# Incremental rotary encoders (IRC)



# Absolute rotary encoders



# Absolute rotary encoders

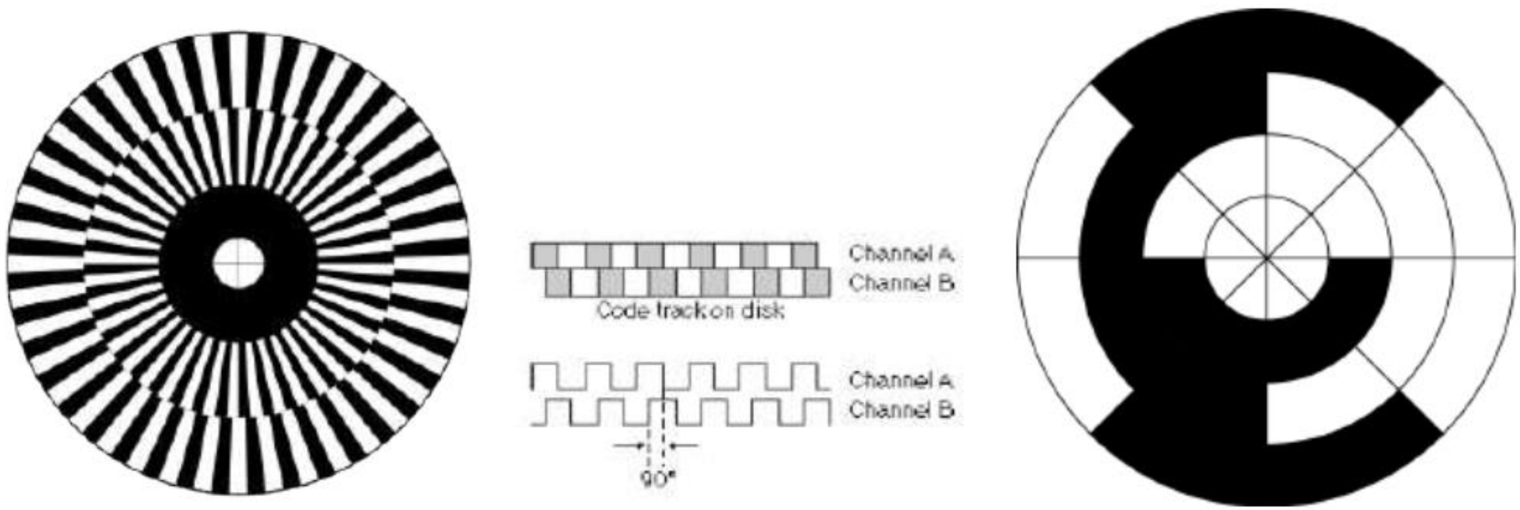
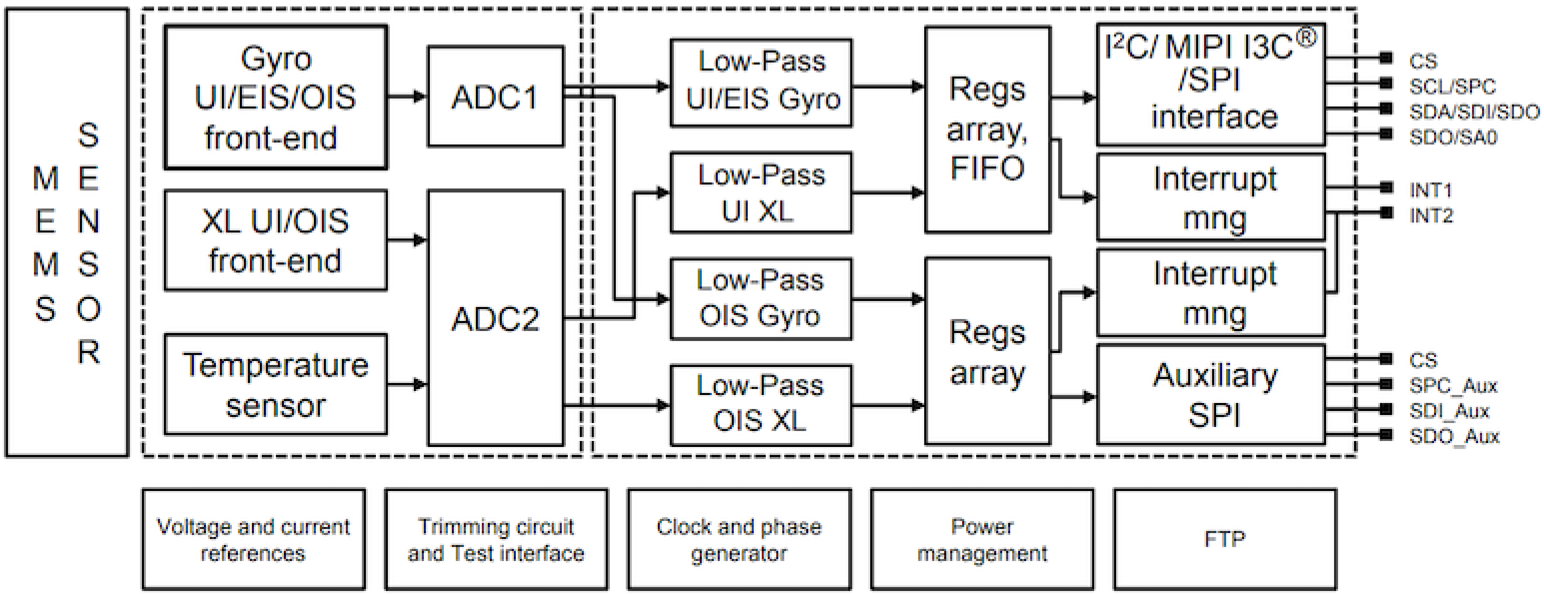


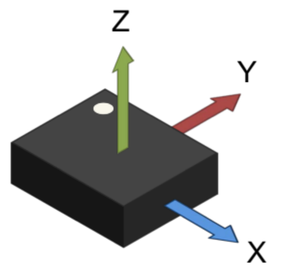

*Figure 7.2.* From left to right: encoder pattern used in a quadrature encoder, resulting sensor signal (forward motion), absolute encoder pattern (gray coding).

# ROBOT – IMU sensor

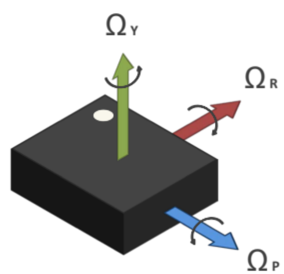

Figure 3. State machine in the LSM6DSV16X

# ROBOT – IMU sensor




Direction of detectable acceleration (top view)




Direction of detectable angular rate (top view)

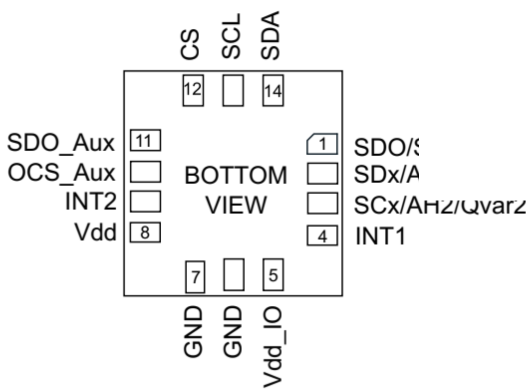



Figure 16. Single-channel mode (XL_DualC_EN = 0)

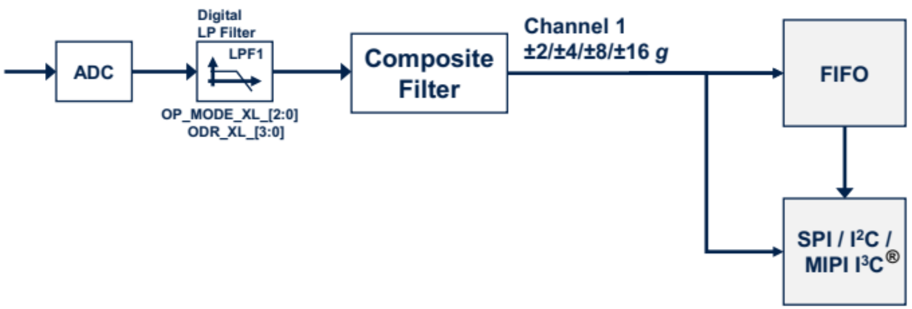
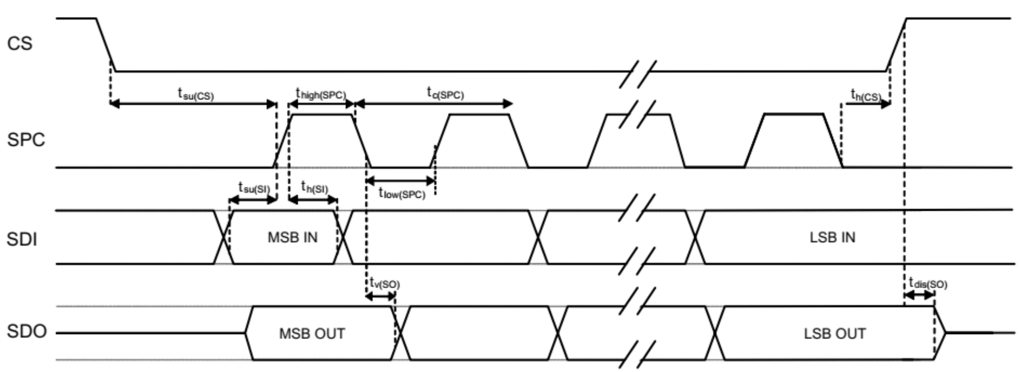



Figure 7. SPI slave timing in mode 0



# ROBOT – IMU sensor

## Digital filter in IMU sensor

Figure 19. Accelerometer UI chain

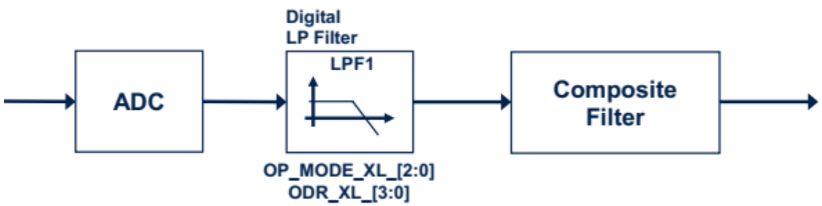



Figure 20. Accelerometer composite filter

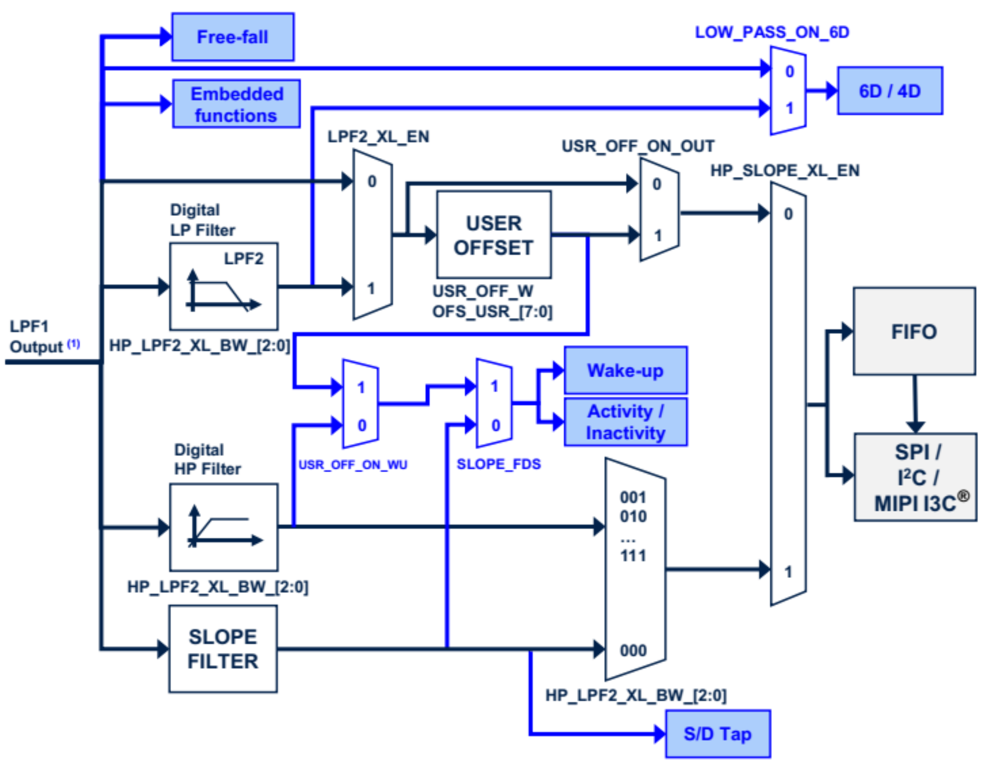
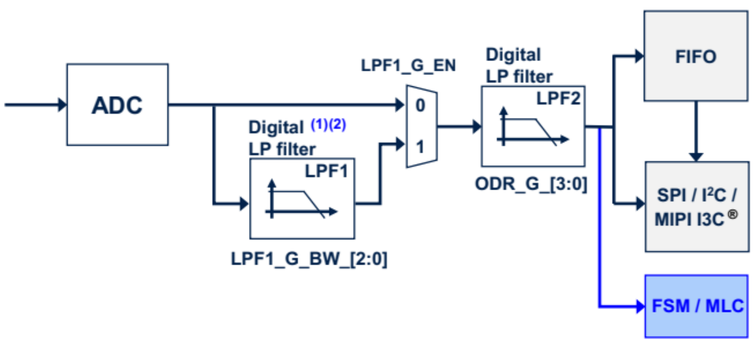
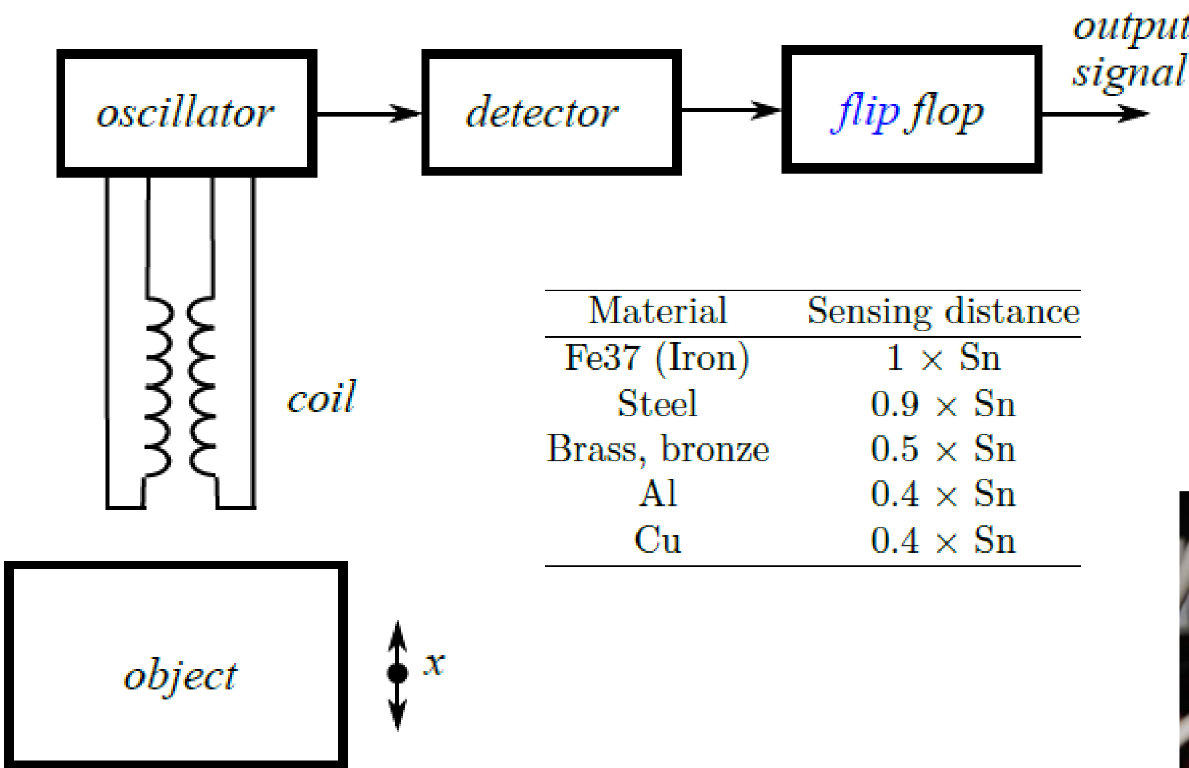



Figure 22. Gyroscope digital chain - mode 1 (UI/EIS) and mode 2

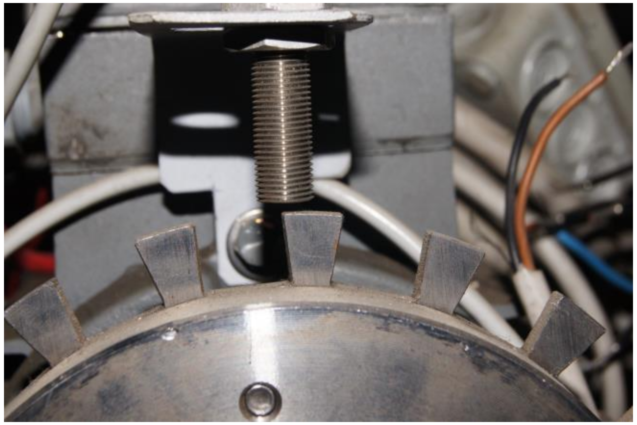


# Inductive proximity sensors



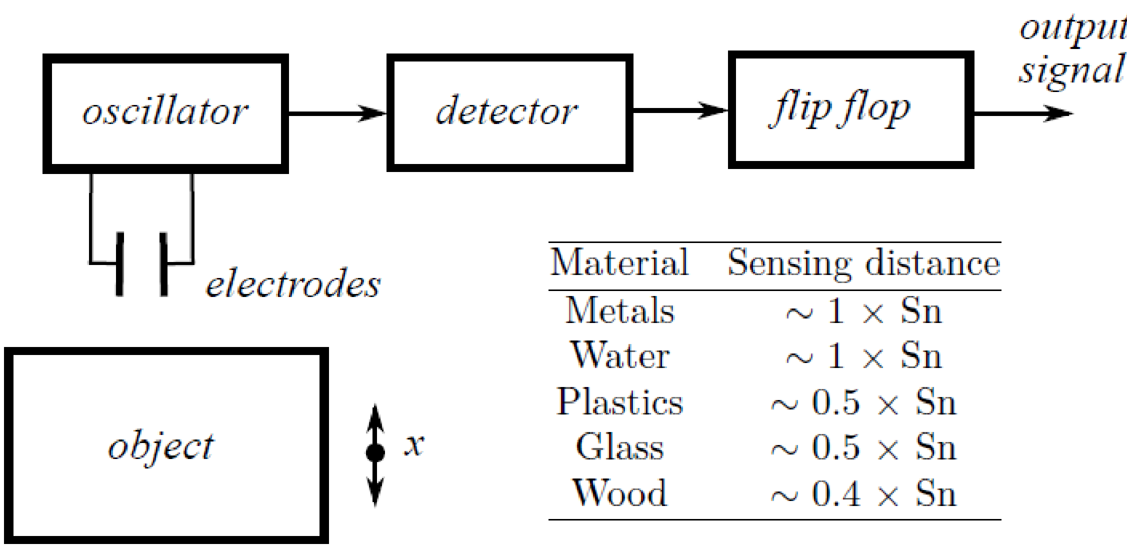

| Material | Sensing distance |
|---|---|
| Fe37 (Iron) | $1 \times Sn$ |
| Steel | $0.9 \times Sn$ |
| Brass, bronze | $0.5 \times Sn$ |
| Al | $0.4 \times Sn$ |
| Cu | $0.4 \times Sn$ |

FIGURE 7.61: Principle of inductive proximity sensor



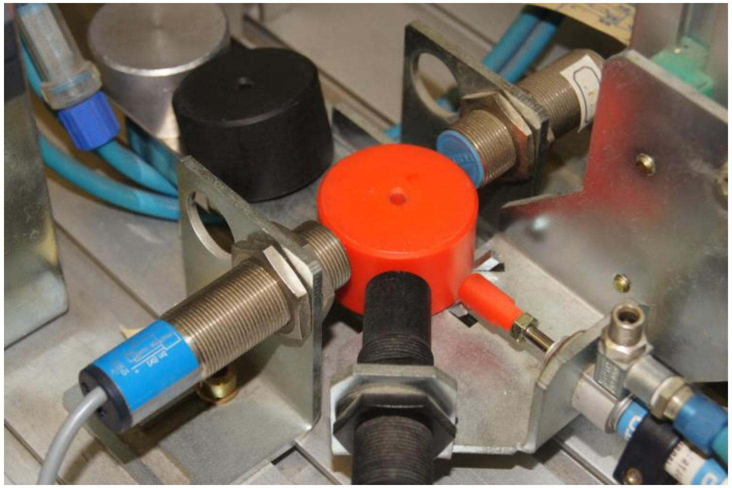
Only work with conductive objects

# Capacitive proximity sensors




| Material | Sensing distance |
|---|---|
| Metals | $\sim 1 \times$ Sn |
| Water | $\sim 1 \times$ Sn |
| Plastics | $\sim 0.5 \times$ Sn |
| Glass | $\sim 0.5 \times$ Sn |
| Wood | $\sim 0.4 \times$ Sn |

FIGURE 7.70: Principle of capacitive proximity sensor



# Optic proximity sensors